To get the dimensions of a matrix (or array) `X` in Python, you can use the `.shape` attribute if `X` is a NumPy array or a Pandas DataFrame. Here's how you can do it for different types of data structures:

1. **If `X` is a NumPy Array**

NumPy arrays have a `.shape` attribute that returns a tuple representing the dimensions of the array.

python Copy

```
import numpy as np

# Example matrix
X = np.array([[1, 2, 3], [4, 5, 6]])

# Get dimensions
dimensions = X.shape
print(dimensions)  # Output: (2, 3)
```

`X.shape` returns `(rows, columns)` for a 2D array.

For a 1D array, it returns `(length,)`.

2. **If `X` is a Pandas DataFrame**

Pandas DataFrames also have a `.shape` attribute that returns the number of rows and columns.

python Copy

```
import pandas as pd

# Example DataFrame
X = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [4, 5, 6],
    'C': [7, 8, 9]
})

# Get dimensions
dimensions = X.shape
print(dimensions)  # Output: (3, 3)
```

`X.shape` returns `(rows, columns)`.

3. **If `X` is a List of Lists**

For a nested list (which can represent a matrix), you can use Python's built-in `len()` function to get the dimensions.

python Copy

```
# Example list of lists
X = [[1, 2, 3], [4, 5, 6]]

# Get dimensions
rows = len(X)
columns = len(X[0]) if rows > 0 else 0
dimensions = (rows, columns)
print(dimensions)  # Output: (2, 3)
```

`len(X)` gives the number of rows.

`len(X[0])` gives the number of columns (assuming all rows have the same length).

4. **If `X` is a SciPy Sparse Matrix**

For sparse matrices (e.g., from `scipy.sparse`), you can also use the `.shape` attribute.

python Copy

```
from scipy.sparse import csr_matrix

# Example sparse matrix
X = csr_matrix([[1, 0, 0], [0, 2, 0]])

# Get dimensions
dimensions = X.shape
print(dimensions)  # Output: (2, 3)
```

Summary of Methods:

Data Structure	Method to Get Dimensions	Example Output
NumPy Array	X.shape	(2, 3)
Pandas DataFrame	X.shape	(3, 3)
List of Lists	len(X) , len(X[0])	(2, 3)
SciPy Sparse Matrix	X.shape	(2, 3)